

Linear combinations and spans

Let V be a vector space (over F) and A a subset of V , then

$$\begin{aligned}\text{Span}(A) &= \text{'smallest subspace of } V \text{ containing } A\text{' } \\ &= \cap \{S \in P(V) \mid S \text{ is a subspace of } V \text{ and } A \subseteq S\}\end{aligned}$$

The latter has a simple description in terms of 'linear combinations', which we shall use often.

Definition 32. Let V be a vector space (over F), then a vector $w \in V$ is a linear combination of vectors $v_1, v_2, \dots, v_k$ if there exists scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ such that $w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k$.

Examples Consider $\mathbb{R}^3$ with usual addition and scalar multiplication

(1) Vector $(3, 1, 4)$ is a linear combination of vectors $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$

since $(3, 1, 4) = 3(1, 0, 0) + 1 \cdot (0, 1, 0) + 4(0, 0, 1)$.

(2) Vector $(3, 1, 4)$ is not a linear combination of vectors $(2, 0, 1)$ and $(3, 0, 0)$ for if it were we would have scalars $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $(3, 1, 4) = \lambda_1(2, 0, 1) + \lambda_2(3, 0, 0) = (2\lambda_1 + 3\lambda_2, 0, \lambda_1)$, notice that the LHS has 1 as its second component and the RHS 0 as its second component and so the two cannot be equal for any $\lambda_1, \lambda_2 \in \mathbb{R}$.

(3) Is vector $(1, 4, 7)$ a linear combination of vectors $(0, 4, 4)$, $(2, 1, 5)$ and $(-1, -1, -3)$? This is equivalent to asking if there exists $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ such that $(1, 4, 7) = \lambda_1(0, 4, 4) + \lambda_2(2, 1, 5) + \lambda_3(-1, -1, -3)$, that is, are there $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ satisfying the following system of linear equations:

$$\begin{aligned}2\lambda_2 - \lambda_3 &= 1 \\ 4\lambda_1 + \lambda_2 - \lambda_3 &= 4 \\ 4\lambda_1 + 5\lambda_2 - 3\lambda_3 &= 7\end{aligned}$$

$$\begin{aligned}\left(\begin{array}{ccc|c} 0 & 2 & -1 & 1 \\ 4 & 1 & -1 & 4 \\ 4 & 5 & -3 & 7 \end{array}\right) &\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 4 & 1 & -1 & 4 \\ 0 & 2 & -1 & 1 \\ 4 & 5 & -3 & 7 \end{array}\right) \\ &\xrightarrow{R_3 \rightarrow R_3 - R_1} \left(\begin{array}{ccc|c} 4 & 1 & -1 & 4 \\ 0 & 2 & -1 & 1 \\ 0 & 4 & -2 & 3 \end{array}\right) \\ &\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left(\begin{array}{ccc|c} 4 & 1 & -1 & 4 \\ 0 & 2 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right)\end{aligned}$$

The above linear system is inconsistent and so $(1, 4, 7)$ is not a linear combination of vectors $(0, 4, 4)$, $(2, 1, 5)$ and $(-1, -1, -3)$. What about $(1, 4, 6)$? Is it a linear combination of vectors $(0, 4, 4)$, $(2, 1, 5)$ and $(-1, -1, -3)$?

(4) Let A be an $m \times n$ matrix and $\mathbf{v}$ an n -column vector, then vector $\mathbf{w} = A\mathbf{v}$ is a linear combination of columns of matrix A .

Notice that if we let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ be columns of matrix A and $\mathbf{v} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}$, then $\mathbf{w} = A\mathbf{v} =$

$$(\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n) \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix} = \lambda_1 \mathbf{c}_1 + \lambda_2 \mathbf{c}_2 + \dots + \lambda_n \mathbf{c}_n.$$

Description of 'span' in terms of linear combinations.

Theorem 13. Let V be a vector space (over F) and A a subset of V . If $A = \emptyset$, then $\text{Span}(A) = \{\mathbf{0}\}$ and if $A \neq \emptyset$, then $\text{Span}(A) = \{w \in V \mid w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \text{ where } v_1, v_2, \dots, v_n \in A, \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R} \text{ and } n \in \mathbb{N}\}$.

Proof. If $A = \emptyset$, then the smallest subspace of V containing A is the trivial subspace $\{\mathbf{0}\}$ and so $\text{Span}(A) = \{\mathbf{0}\}$. Now let $A \neq \emptyset$ we show that $\text{Span}(A) = \{w \in V \mid w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \text{ where } v_1, v_2, \dots, v_n \in A, \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R} \text{ and } n \in \mathbb{N}\}$ is the smallest subspace of V containing A .

We first check that it is a subspace of V :

Notice that $\mathbf{0} \in \text{Span}(A)$, since if we take $v \in A$, we have that $0v = \mathbf{0} \in \text{Span}(A)$. Now if $w, w' \in \text{Span}(A)$ we then have that $w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n$ and $w' = \lambda'_1 v'_1 + \lambda'_2 v'_2 + \dots + \lambda'_m v'_m$ and so $w + w' = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n + \lambda'_1 v'_1 + \lambda'_2 v'_2 + \dots + \lambda'_m v'_m \in \text{Span}(A)$ and if $\mu \in F$ we have $\mu w = (\mu \lambda_1) v_1 + (\mu \lambda_2) v_2 + \dots + (\mu \lambda_n) v_n \in \text{Span}(A)$ and so $\text{Span}(A)$ is a subspace of V .

We now check that it is the smallest subspace of V containing A :

Let W be a subspace of V such that $A \subseteq W$ and $w \in \text{Span}(A)$, then $w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n$ with $v_1, v_2, \dots, v_n \in A \subseteq W$, since W is a subspace we have $\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n \in W$ and so $w = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \in W$, that is $\text{Span}(A) \subseteq W$. $\square$

Let V be a vector space (over F), in particular if $A = \{v_1, v_2, \dots, v_n\}$, then $\text{Span}(\{v_1, v_2, \dots, v_n\}) = \{\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \mid \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}\}$.

Examples: (1) Consider $V = \mathbb{R}^3$ with usual addition and scalar multiplication

$$\begin{aligned} \text{Span}(\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}) &= \{\lambda_1(1, 0, 0) + \lambda_2(0, 1, 0) + \lambda_3(0, 0, 1) \mid \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}\} \\ &= \{(\lambda_1, \lambda_2, \lambda_3) \mid \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}\} \\ &= \mathbb{R}^3 \end{aligned}$$

(2) Consider $V = \mathbb{R}^{\mathbb{R}}$ the set of all real-valued functions with domain $\mathbb{R}$ with usual addition and scalar multiplication, that is, if $f, g \in V$ and $\lambda \in \mathbb{R}$, then $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$.

Let $f_0, f_1, \dots, f_n \in V$ be given by $f_0(x) = 1, f_1(x) = x, \dots, f_n(x) = x^n$ we have that

$$\begin{aligned} \text{Span}(\{f_0, f_1, \dots, f_n\}) &= \{\lambda_0 f_0 + \lambda_1 f_1 + \dots + \lambda_n f_n \mid \lambda_0, \lambda_1, \dots, \lambda_n \in \mathbb{R}\} \\ &= \{\lambda_0 + \lambda_1 x + \dots + \lambda_n x^n \mid \lambda_0, \lambda_1, \dots, \lambda_n \in \mathbb{R}\} \\ &= P_n = \text{'subspace of polynomials of degree at most } n' \end{aligned}$$

(3) Let P_2 be polynomials of degree at most 2 with coefficient-wise addition and scalar multiplication. Is polynomial $p = 1 + x^2$ in the span of set of polynomials $p_1 = 1 - x^2, p_2 = x + x^2$ and $p_3 = 1 + x - x^2$, that is, is $p \in \text{Span}(\{p_1, p_2, p_3\})$? This is equivalent to asking if there exists scalars $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ such that $p = \lambda_1 p_1 + \lambda_2 p_2 + \lambda_3 p_3$, that is

$$\begin{aligned} 1 + x^2 &= \lambda_1(1 - x^2) + \lambda_2(x + x^2) + \lambda_3(1 + x - x^2) \\ &= (\lambda_1 + \lambda_3) + (\lambda_2 + \lambda_3)x + (-\lambda_1 + \lambda_2 - \lambda_3)x^2 \end{aligned}$$

and so it is a question of whether or not the system of linear equations below has a solution (complete the details).

$$\lambda_1 + \lambda_3 = 1$$

$$\lambda_2 + \lambda_3 = 0$$

$$-\lambda_1 + \lambda_2 - \lambda_3 = 1$$